BRIEF-005: Zero-Knowledge Clinical Reasoning in Confidential Enclaves

PHI-Safe Pipelines & Cryptographic Isolation 2025

STAGE-X.SPACE TECHNICAL ARCHITECTURE BRIEF
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EXECUTIVE SUMMARY:
Clinical reasoning systems operating inside hardware-isolated AMD SEV-SNP confidential enclaves ensuring patient health information is cryptographically inaccessible to host OS.

KEY ARCHITECTURE PROOFS:
1. Hardware Attestation: Remote attestation of hardware root of trust before key exchange.
2. Ephemeral Isolation: Session memory keys destroyed upon completion of reasoning pass.
3. Verification Pipeline: Deterministic formulary cross-checks with citation chains.
4. Regulatory Compliance: Full HIPAA and GDPR Article 9 technical isolation proofs.

FIELD MEASUREMENTS:
- Cryptographic Boundary Leakage: 0 bytes across continuous automated penetration tests
- Verification Pass Latency: < 120ms P95 per clinical interaction
- Formulary Alignment Parity: 100% adherence to peer-reviewed clinical protocols